

TAMIL NADU STATE BOARD - STANDARD 10 MATHEMATICS

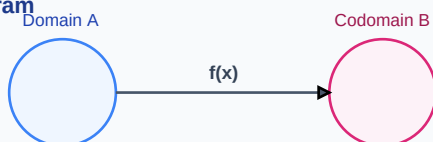
CHAPTER 4: GEOMETRY

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 10 English Medium
Subject:	Mathematics
Chapter Name:	Geometry

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram

Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Similar Triangles	Triangles having the same shape but not necessarily the same size. Corresponding angles are equal and sides are proportional.
Thales Theorem	If a line is drawn parallel to one side of a triangle to intersect the other two sides, it divides them in the same ratio.
Tangent to Circle	A straight line that touches the circle at exactly one point, perpendicular to the radius at that contact point.
Cyclic Quadrilateral	A quadrilateral whose all four vertices lie on the circumference of a single circle.

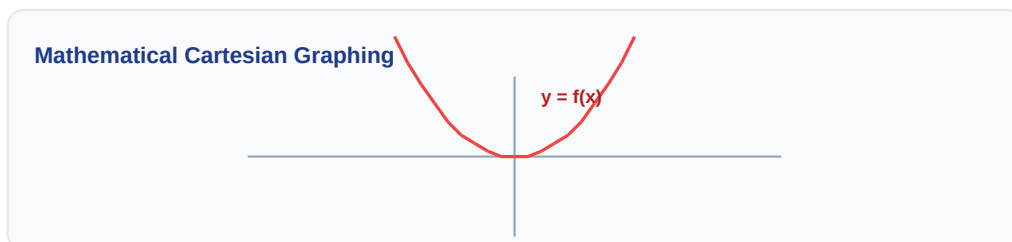
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Basic Proportionality Theorem (BPT): $AD / DB = AE / EC$
- Angle Bisector Theorem: $BD / DC = AB / AC$
- Alternate Segment Theorem: Angle between chord and tangent equals angle in alternate segment
- Intersecting Chords Rule: $PA * PB = PC * PD$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: In a triangle ABC, D and E are points on AB and AC such that $DE \parallel BC$. If $AD=2\text{cm}$, $DB=3\text{cm}$, and $AE=4\text{cm}$, find EC.

Step-by-step Solution:

By Thales (BPT) theorem: $AD / DB = AE / EC$.

Substitute values: $2 / 3 = 4 / EC$.

Cross multiply: $2 * EC = 12 \Rightarrow EC = 6 \text{ cm}$.

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: State and prove the Angle Bisector Theorem (ABT).

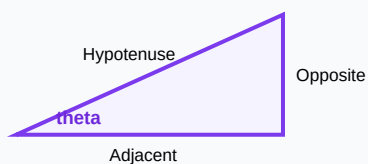
Analytical Solution Strategy:

ABT states that the internal bisector of an angle of a triangle divides the opposite side internally in the ratio of the corresponding sides containing the angle.

Construction: Draw CE parallel to AB meeting AD produced at E.

Proof utilizes similar triangles ABD and ECD to derive $BD / DC = AB / AC$.

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): State Thales Theorem (Basic Proportionality Theorem).

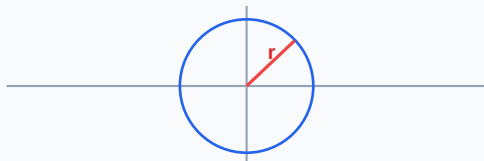
Q2 (2-Mark): In a circle of radius 5 cm, find the length of tangent from a point 13 cm away from the center.

Q3 (2-Mark): Show that corresponding altitudes of similar triangles are proportional to their bases.

Q4 (5-Mark): State the Alternate Segment Theorem.

Q5 (5-Mark): A cyclic quadrilateral ABCD has angle A = 75 degrees. Find angle C.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Navigation

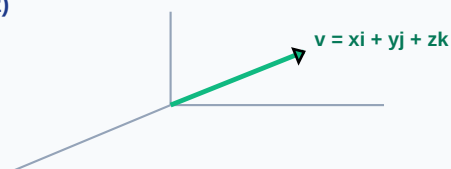
Similarity principles allow cartographers to draw scaled maps representing actual terrains with accurate geometry.

Application Case: Bridge Truss Analysis

Triangulation models determine structural load distributions and load thresholds for safe civil constructions.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

